

Proof of RLS and (1+1) EA for Function F3

Theorem:

The number of fitness evaluations of RLS and (1+1) EA to reach an optimal search point for the function F3 is $O(n \log n)$ with probability $\Omega(1)$.

Problem Setup:

Function F3: A Linear Function with Harmonic Weights $f: \{0,1\}^n \rightarrow \mathbb{R}, x$

$$\mapsto \sum_{i=1}^n i \cdot x_i$$

Optimal solution: $x^* = 1^n$

All weights are positive, so this is a monotone function (flipping 0→1 always improves fitness)

Algorithm Definitions:

RLS (Randomized Local Search):

- Each iteration: flip exactly 1 bit chosen uniformly at random
- Accept if fitness doesn't decrease

(1+1) EA:

- Each iteration: flip each bit independently with probability $1/n$
- Accept if fitness doesn't decrease

Proof for RLS:

Step 1: Expected Runtime Analysis

Starting from any initial solution, suppose we currently have k bits set to 1. We need to analyze how long it takes to set all remaining $(n-k)$ zero-bits to 1.

- Flipping a $0 \rightarrow 1$ always improves fitness (accepted)
- Flipping a $1 \rightarrow 0$ always decreases fitness (rejected)

When we have $(n-k)$ zero-bits remaining:

- Probability of selecting a 0-bit: $(n-k)/n$
- Expected time until next improvement: $n/(n-k)$

Total expected runtime:

$$E[T] = \sum_{k=0}^{n-1} n/(n-k) = \sum_{j=1}^n n/j = n \sum_{j=1}^n 1/j = n \cdot H_n$$

where $H_n = \sum_{i=1}^n 1/i$ is the n -th harmonic number.

Since $H_n = \Theta(\log n)$, we have: $E[T] = \Theta(n \log n)$

Step 2: Apply Markov's Inequality:

Markov's Inequality states: For a non-negative random variable X and $a > 0$:

$$P(X \geq a) \leq E[X] / a$$

Equivalently:

$$P(X < a) \geq 1 - (E[X] / a)$$

Let's choose $a = 2 \cdot E[T] = 2nH_n = O(n \log n)$:

$$P(T < 2E[T]) \geq 1 - (E[T] / 2E[T]) = 1 - (1/2) = 1/2$$

Conclusion for RLS:

With probability at least $1/2 = \Omega(1)$, RLS finds the optimum in at most $2E[T] = O(n \log n)$ fitness evaluations.

Proof for (1+1) EA

Step 1: Expected Runtime Analysis

When we have k zero-bits remaining, an improvement occurs when:

- At least one 0-bit flips to 1, AND
- No 1-bits flip to 0

Lower bound on improvement probability:

The probability of flipping exactly one specific 0-bit (and no other bits):

$$p = 1/n \cdot (1 - (1/n)^{n-1}) \geq 1/n \cdot 1/e = 1/en$$

With k zero-bits, probability of improving (at least one improves):

$$p_{\text{improve}} \geq k/en$$

Expected time until improvement: $E[T_k] \leq en/k$

Total expected runtime:

$$E[T] \leq \sum_{k=1}^n en/k = en \cdot \sum_{k=1}^n 1/k = en \cdot H_n = \Theta(n \log n)$$

Step 2: Apply Markov's Inequality

Same as RLS, choose $a = 2E[T] = O(n \log n)$:

$$P(T < 2E[T]) \geq 1 - (E[T]/2E[T]) = 1/2$$

Conclusion for (1+1) EA:

With probability at least $1/2 = \Omega(1)$, (1+1) EA finds the optimum in at most $O(n \log n)$ fitness evaluations.

Conclusion:

Both RLS and (1+1) EA achieve:

- Expected runtime: $E[T] = \Theta(n \log n)$

- With probability $\Omega(1)$ (specifically $\geq 1/2$), runtime is $O(n \log n)$

This is proven using:

- 1: Computing expected runtime using harmonic sum
- 2: Applying Markov's inequality to bound tail probability